

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_failure_replay.
- Best TSPLIB ATSP holdout gap: atsp_failure_replay_compression.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_181441_o
- started_at_local: 2026-05-13 18:14:41
- finished_at_local: 2026-05-13 18:23:02
- duration_hhmm: 00:08
- duration_seconds: 500.898
- seed_offset: 14000
- replicate_label: o
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.192839	0.009004	0.100922	0.844537	0.58	0.012419
atsp_random_replay	random	score_only	False	0.192888	0.008454	0.100671	0.910159	0.58	0.011694
atsp_failure_replay	failure	score_only	False	0.190811	0.009004	0.099908	0.869649	0.58	0.018694
atsp_failure_replay_compression	failure	novelty_gate	True	0.186239	0.030781	0.10851	0.0	0.78	0.0

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.192839, synthetic 0.009004, combined 0.100922.

- Accepted-epoch count 4, mean accepted code novelty 0.844537, and final complexity 0.58.
- Adaptation efficiency 0.012419 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.192839 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.003915, ry=0.109416.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.192888, synthetic 0.008454, combined 0.100671.
- Accepted-epoch count 4, mean accepted code novelty 0.910159, and final complexity 0.58.
- Adaptation efficiency 0.011694 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.192888 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.004982, ry=0.109416.
- Panel synthetic_holdout mean gap 0.008454 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.190811, synthetic 0.009004, combined 0.099908.
- Accepted-epoch count 3, mean accepted code novelty 0.869649, and final complexity 0.58.
- Adaptation efficiency 0.018694 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.190811 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.004093, ry=0.101997.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.186239, synthetic 0.030781, combined 0.10851.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.

- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.186239 across 4 instances; family means: ft=0.303548, ftv=0.279603, p=0.001423, ry=0.16038.
- Panel synthetic_holdout mean gap 0.030781 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.037511, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1973, "family": "ftv", "name": "ftv38", "optimality_gap": 0.289542}, {"best_known_cost": 1286, "cost": 1600, "family": "ftv", "name": "ftv33", "optimality_gap": 0.244168}, {"best_known_cost": 1473, "cost": 1828, "family": "ftv", "name": "ftv35", "optimality_gap": 0.241005}]

Judge Appendix

Summary of Results

Condition	Transfer Gap (TSPLIB)	Synthetic Holdout Gap	TSPLIB Gap	Code Novelty (mean)	Complexity	Replay Mode	Notes
atsp_no_replay	0.1009	0.0090	0.1928	0.8445	0.58	none	Baseline performance, moderate transfer gap and balanced complexity
atsp_random_replay	0.1007	**0.0085**	0.1929	0.9102 (highest)	0.58	random	Best synthetic gap (transfer to synthetic data improved), code novelty highest
atsp_failure_replay	**0.0999 (best)**	0.0090	**0.1908 (best)**	0.8696	0.58	failure	Best transfer gaps (TSPLIB and transfer), lower code novelty than random replay
atsp_failure_replay_compression	0.1085 (worst)	0.0308 (worst)	0.1862 (best)	0.0 (lowest)	0.78 (highest)	failure + compression	Worst transfer and synthetic gaps, lowest code novelty, highest complexity, compression pressure on

Interpretation

- **Transfer performance (TSPLIB and synthetic holdouts):**
- *atsp_failure_replay* condition achieves the **best transfer gaps on TSPLIB (0.1908) and combined transfer measure (0.0999)**, indicating more effective transfer learning across diverse ATSP instances.
- *atsp_random_replay* shows the best synthetic holdout gap (0.0085) but slightly worse TSPLIB gap compared to failure replay.
- *atsp_failure_replay_compression* performs worse on transfer and synthetic holdouts despite achieving a slightly better TSPLIB gap (0.1862), but with higher complexity and zero code novelty, indicating possible overfitting or stagnation.
- **Code Novelty vs Transfer:**
- *atsp_random_replay* maintains the highest average code novelty (~0.91) with similar transfer gap values to baseline no replay, demonstrating that increased lexical novelty does **not** translate to improved transfer gaps.
- *atsp_failure_replay* improves transfer while slightly reducing code novelty relative to random replay, suggesting that better transfer is associated with more focused or effective replay (failure-based) rather than raw novelty.

- *Compression-aware replay* leads to **lowest code novelty** and worst transfer results, despite higher complexity, implying that compression pressure negatively affects both novelty and transfer.
- **Replay Modes:**
- **No replay (baseline)** has modest transfer and novelty.
- **Random replay** improves synthetic transfer measure and code novelty but not TSPLIB transfer.
- **Failure replay** improves transfer metrics (TSPLIB and transfer gaps) most effectively, supporting targeted replay over random or none.
- **Failure replay with compression pressure** degrades transfer and novelty, despite high complexity, indicating complexity increase does not guarantee better performance.

Conservative Conclusions

- The condition **atsp_failure_replay** yields the **best overall transfer performance** (lowest TSPLIB and transfer gaps), validating failure replay as the most effective replay mode to enhance generalization.
- Improvement in transfer with failure replay occurs **despite a small decrease in code novelty**, indicating that lexical novelty alone does not drive algorithmic improvements in transfer.
- Random replay increases code novelty but does **not** significantly improve or surpass failure replay in transfer quality.
- Compression-aware replay suppresses code novelty and harms transfer despite increased complexity, suggesting compression pressure is detrimental under current conditions.
- No replay baseline performs reasonably but is outperformed by replay-based methods on transfer gaps.

Recommendations

- Prioritize **failure replay** mechanisms to achieve improved transfer to held-out TSPLIB and synthetic ATSP instances.
- Do not equate higher code novelty or complexity to better transfer unless accompanied by reduced optimality gaps.
- Avoid compression pressure in failure replay without further tuning due to negative impact on transfer and novelty.
- Further study is needed to isolate why compression-aware replay increases complexity but harms transfer and novelty.